

COMPUTER NETWORKS LAB

3-Router OSPF Network with VLANs, DHCP & DNS

1. Introduction

This Project Report documents the design, configuration, and verification of a 3-router OSPF network built in Cisco Packet Tracer. The network incorporates VLANs at each site, router-based DHCP servers, a centralized DNS server, and route summarization techniques. The objective is to simulate a realistic multi-site network and verify correct routing, addressing, and name resolution.

The key learning outcomes of this lab include:

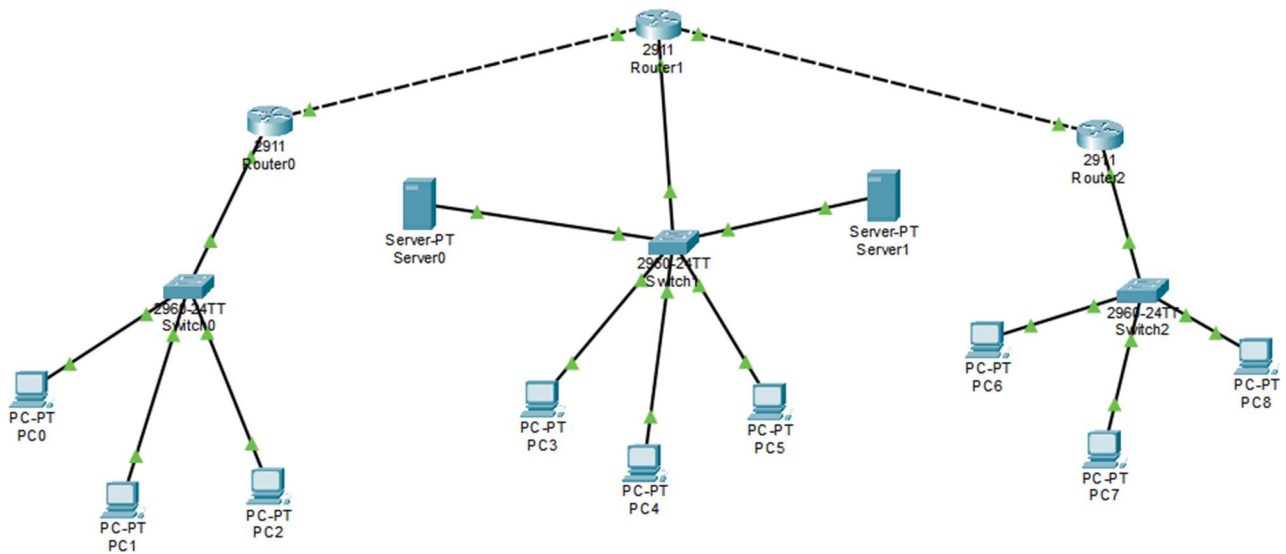
- Configuring OSPF (Open Shortest Path First) across multiple routers
- Setting up VLANs and assigning access ports on switches
- Deploying DHCP pools on routers for automatic IP assignment
- Configuring a DNS server for hostname-to-IP resolution
- Verifying OSPF neighbor relationships and LSA propagation
- Practicing route summarization for efficient routing tables

2. Network Topology

The network consists of three routers (R1, R2, R3) connected in a linear chain, with each router serving a separate LAN segment via a managed switch and end-user PCs. R2 acts as the central hub, connecting to both R1 and R3. A DHCP/DNS server is attached to the VLAN20 segment managed by S2.

2.1 Topology Diagram

Screenshot: Network Topology Diagram (Cisco Packet Tracer)



2.2 Port & Connection Table

Device	Port	Connected To	VLAN / Subnet	Notes
R1	G0/0	R2 G0/0	10.0.12.0/24	OSPF link
R1	G0/1	S1 Fa0/1	192.168.10.0/24	VLAN10
S1	Fa0/2–4	PC0–PC2	VLAN10	Access ports
R2	G0/0	R1 G0/0	10.0.12.0/24	OSPF link
R2	G0/1	R3 G0/0	10.0.23.0/24	OSPF link
R2	G0/2	S2 Fa0/1	192.168.20.0/24	VLAN20, DHCP
S2	Fa0/2–4	PC3–PC5	VLAN20	Access ports
S2	Fa0/5	DHCP/DNS Server	VLAN20	Straight-through cable
R3	G0/0	R2 G0/1	10.0.23.0/24	OSPF link
R3	G0/1	S3 Fa0/1	192.168.30.0/24	VLAN30
S3	Fa0/2–4	PC6–PC8	VLAN30	Access ports

3. Router Configurations

Each router is configured with IP addressing on all interfaces, OSPF Area 0 routing, and a DHCP pool serving its local LAN segment.

3.1 Router R1

R1 connects to R2 via G0/0 and serves VLAN10 (192.168.10.0/24) via G0/1. It also acts as the DHCP server for VLAN10 clients.

```
enable
configure terminal
hostname R1

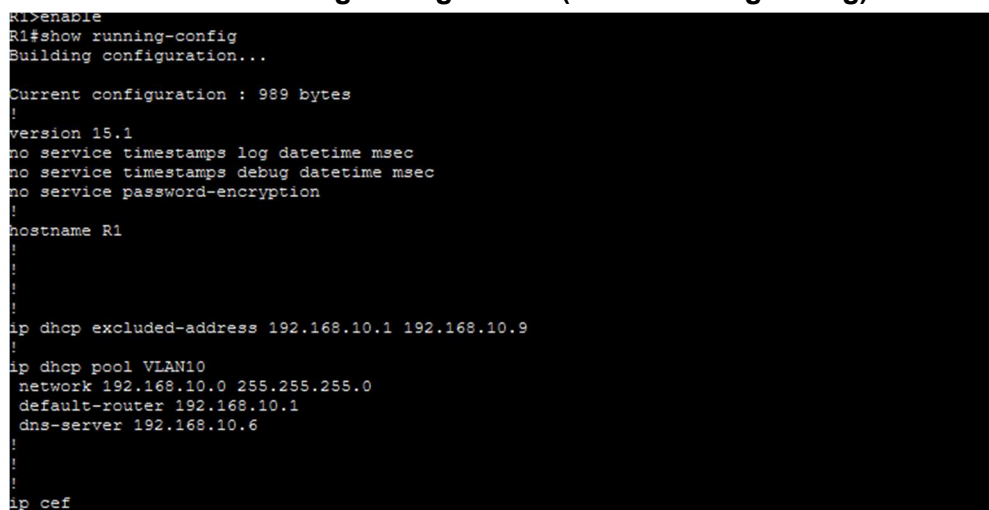
interface GigabitEthernet0/0
 ip address 10.0.12.1 255.255.255.0
 no shutdown

interface GigabitEthernet0/1
 ip address 192.168.10.1 255.255.255.0
 no shutdown

router ospf 1
 log-adjacency-changes
 network 10.0.12.0 0.0.0.255 area 0
 network 192.168.10.0 0.0.0.255 area 0

ip dhcp excluded-address 192.168.10.1 192.168.10.9
ip dhcp pool VLAN10
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 192.168.10.6
```

Screenshot: R1 Running Configuration (show running-config)

A screenshot of a terminal window showing the running configuration of Router R1. The output of the 'show running-config' command is displayed, showing the configuration for the router, including interfaces, OSPF, and DHCP. The configuration matches the commands shown in the previous block.

```
R1>enable
R1#show running-config
Building configuration...

Current configuration : 989 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
!
!
!
!
ip dhcp excluded-address 192.168.10.1 192.168.10.9
!
ip dhcp pool VLAN10
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 192.168.10.6
!
!
!
ip cef
```

```
interface GigabitEthernet0/0
 ip address 10.0.12.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 192.168.10.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
 network 192.168.10.0 0.0.0.255 area 0
 network 10.0.12.0 0.0.0.255 area 0
!
ip classless
!
ip flow-export version 9
```

3.2 Router R2

R2 is the central router connecting to both R1 (G0/0) and R3 (G0/1). It also serves VLAN20 (192.168.20.0/24) via G0/2 and provides DHCP to VLAN20 clients.

```
enable
configure terminal
hostname R2

interface GigabitEthernet0/0
 ip address 10.0.12.2 255.255.255.0
 no shutdown

interface GigabitEthernet0/1
 ip address 10.0.23.1 255.255.255.0
 no shutdown

interface GigabitEthernet0/2
 ip address 192.168.20.1 255.255.255.0
 no shutdown

router ospf 1
 log-adjacency-changes
```

```
network 10.0.12.0 0.0.0.255 area 0
network 10.0.23.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0

ip dhcp excluded-address 192.168.20.1 192.168.20.9
ip dhcp pool VLAN20
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 192.168.20.6
```

Screenshot: R2 Running Configuration (show running-config)

```
Building configuration...

Current configuration : 1036 bytes
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption

hostname R2

ip dhcp excluded-address 192.168.20.1 192.168.20.9

ip dhcp pool VLAN20
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 192.168.20.6
```

```
interface GigabitEthernet0/0
ip address 10.0.12.2 255.255.255.0
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 10.0.23.1 255.255.255.0
duplex auto
speed auto
!
interface GigabitEthernet0/2
ip address 192.168.20.1 255.255.255.0
duplex auto
speed auto
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 10.0.12.0 0.0.0.255 area 0
network 10.0.23.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
!
ip classless
!
ip flow-export version 9
!
```

3.3 Router R3

R3 connects to R2 via G0/0 and serves VLAN30 (192.168.30.0/24) via G0/1. It provides DHCP service for VLAN30 clients.

```
enable
configure terminal
hostname R3

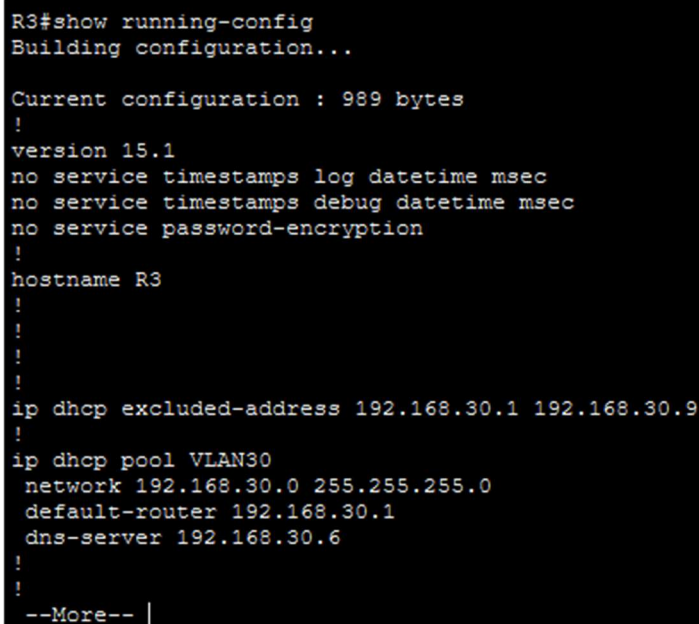
interface GigabitEthernet0/0
 ip address 10.0.23.2 255.255.255.0
 no shutdown

interface GigabitEthernet0/1
 ip address 192.168.30.1 255.255.255.0
 no shutdown

router ospf 1
 log-adjacency-changes
 network 10.0.23.0 0.0.0.255 area 0
 network 192.168.30.0 0.0.0.255 area 0

ip dhcp excluded-address 192.168.30.1 192.168.30.9
ip dhcp pool VLAN30
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 192.168.30.6
```

Screenshot: R3 Running Configuration (show running-config)

A screenshot of a terminal window showing the running configuration of Router R3. The output of the 'show running-config' command is displayed, including system version, service settings, hostname, interfaces, OSPF configuration, and DHCP pool configuration. The terminal text is as follows:

```
R3#show running-config
Building configuration...

Current configuration : 989 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R3
!
!
!
!
ip dhcp excluded-address 192.168.30.1 192.168.30.9
!
ip dhcp pool VLAN30
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 192.168.30.6
!
!
--More-- |
```

```
interface GigabitEthernet0/0
 ip address 10.0.23.2 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 192.168.30.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
 network 10.0.23.0 0.0.0.255 area 0
 network 192.168.30.0 0.0.0.255 area 0
!
ip classless
!
ip flow-export version 9
!
```

4. Switch Configurations

Each switch is configured with its respective VLAN and all access ports are assigned to that VLAN. All ports are brought up with the no shutdown command.

4.1 Switch S1 – VLAN10

```
vlan 10
  name VLAN10

interface fa0/1
  switchport mode access
  switchport access vlan 10
  no shutdown
interface range fa0/2 - 4
  switchport mode access
  switchport access vlan 10
  no shutdown
```

4.2 Switch S2 – VLAN20

```
vlan 20
  name VLAN20

interface fa0/1
  switchport mode access
  switchport access vlan 20
  no shutdown
interface range fa0/2 - 5
  switchport mode access
  switchport access vlan 20
  no shutdown
```

4.3 Switch S3 – VLAN30

```
vlan 30
  name VLAN30

interface fa0/1
  switchport mode access
  switchport access vlan 30
  no shutdown
interface range fa0/2 - 4
  switchport mode access
  switchport access vlan 30
  no shutdown
```

Screenshot: Switch VLAN verification (show vlan brief) on S1, S2, or S3

Switch1

```
Switch>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	VLAN10	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch>
```

Switch2

```
Switch>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
20	VLAN20	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Switch3

```
Switch>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
30	VLAN30	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

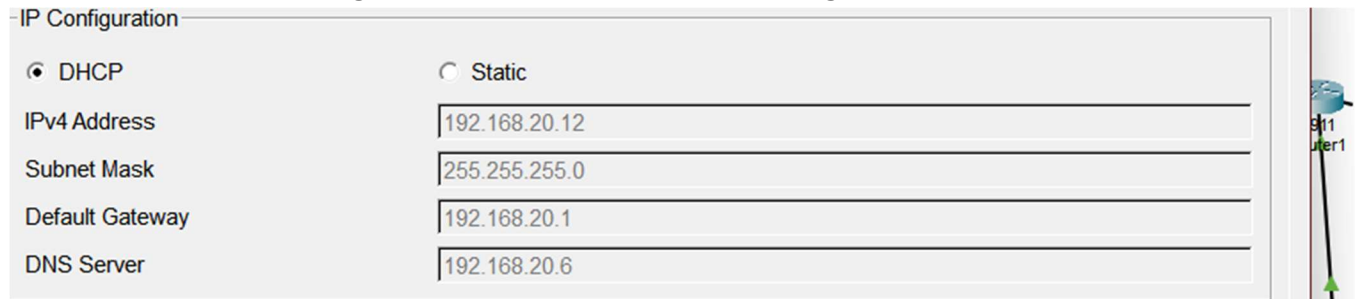
5. DHCP Configuration & Verification

DHCP pools are configured directly on each router. A range of addresses (.1 to .9) is excluded to reserve IPs for static assignments such as the gateway and DNS server. PCs obtain their IP addresses automatically via DHCP.

5.1 DHCP Summary

VLAN	Network	Gateway	DNS Server	Excluded Range
VLAN10	192.168.10.0/24	192.168.10.1	192.168.10.6	.1 – .9
VLAN20	192.168.20.0/24	192.168.20.1	192.168.20.6	.1 – .9
VLAN30	192.168.30.0/24	192.168.30.1	192.168.30.6	.1 – .9

Screenshot: PC obtaining IP via DHCP (Desktop > IP Configuration > DHCP)



Screenshot: show ip dhcp binding on a router

```
R2#show ip dhcp binding
IP address      Client-ID/
Hardware address Lease expiration Type
192.168.20.10   0060.2F6B.0BCE --      Automatic
192.168.20.11   0030.A395.6537 --      Automatic
192.168.20.12   0001.64D3.8983 --      Automatic
R2#
```

6. DNS Server Configuration

A DNS server is connected to Switch S2 on port Fa0/5 with a static IP address of 192.168.20.6. It provides hostname-to-IP resolution for devices within the network.

6.1 DNS Server IP Settings

Parameter	Value
IP Address	192.168.20.6
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.1
Connected Switch Port	S2 Fa0/5

6.2 DNS Hostname Entries

Hostname	IP Address
PC3	192.168.20.11
PC4	192.168.20.12
PC5	192.168.20.13

Screenshot: DNS Server configuration panel in Packet Tracer

DNS Service

☒ On ☐ Off

Resource Records

Name

Type

A Record

Address

Add

Save

Remove

No.	Name	Type	Detail
0	pc3	A Record	192.168.20.11
1	pc4	A Record	192.168.20.12

Screenshot: Successful ping using hostname (e.g., ping PC4)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping pc4

Pinging 192.168.20.12 with 32 bytes of data:

Reply from 192.168.20.12: bytes=32 time=14ms TTL=128
Reply from 192.168.20.12: bytes=32 time=7ms TTL=128
Reply from 192.168.20.12: bytes=32 time=12ms TTL=128
Reply from 192.168.20.12: bytes=32 time=9ms TTL=128

Ping statistics for 192.168.20.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 14ms, Average = 10ms

C:\>
```

7. OSPF Configuration & Verification

OSPF (Open Shortest Path First) is configured in Area 0 across all three routers. All directly connected networks are advertised using the network command with wildcard masks.

7.1 OSPF Neighbor Relationships

After configuration, R1–R2 and R2–R3 form OSPF adjacencies. These can be verified using the show ip ospf neighbor command.

```
R1# show ip ospf neighbor
R2# show ip ospf neighbor
R3# show ip ospf neighbor
```

Screenshot: show ip ospf neighbor output on R1, R2, and R3

```
R2#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
192.168.30.1    1     FULL/DR         00:00:37   10.0.23.2     GigabitEthernet0/1
192.168.10.1    1     FULL/BDR        00:00:30   10.0.12.1     GigabitEthernet0/0
R2#
```

7.2 OSPF Database / LSA Propagation

The OSPF link-state database shows all LSAs flooded across the network. Each router should have a complete view of the entire topology.

```
R1# show ip ospf database
R2# show ip ospf database
R3# show ip ospf database
```

Screenshot: show ip ospf database output**R1**

```
R1>show ip ospf database
      OSPF Router with ID (192.168.10.1) (Process ID 1)

      Router Link States (Area 0)

Link ID        ADV Router    Age          Seq#           Checksum Link count
192.168.10.1   192.168.10.1   1427         0x8000000b    0x00d1dc 2
192.168.20.1   192.168.20.1   1426         0x8000000a    0x002d0e 3
192.168.30.1   192.168.30.1   652          0x80000005    0x005a07 2

      Net Link States (Area 0)

Link ID        ADV Router    Age          Seq#           Checksum
10.0.12.2      192.168.20.1   1426         0x80000002    0x00bc8e
10.0.23.2      192.168.30.1   652          0x80000003    0x00cf24
R1>
```

R2

```
R2#show ip ospf database
      OSPF Router with ID (192.168.20.1) (Process ID 1)

      Router Link States (Area 0)

Link ID        ADV Router    Age          Seq#           Checksum Link count
192.168.20.1   192.168.20.1   1385         0x8000000a    0x002d0e 3
192.168.10.1   192.168.10.1   1386         0x8000000b    0x00d1dc 2
192.168.30.1   192.168.30.1   611          0x80000005    0x005a07 2

      Net Link States (Area 0)

Link ID        ADV Router    Age          Seq#           Checksum
10.0.12.2      192.168.20.1   1385         0x80000002    0x00bc8e
10.0.23.2      192.168.30.1   611          0x80000003    0x00cf24
R2#
```

R3

```
R3#
R3#show ip ospf database
      OSPF Router with ID (192.168.30.1) (Process ID 1)

      Router Link States (Area 0)

Link ID        ADV Router    Age          Seq#           Checksum Link count
192.168.30.1   192.168.30.1   689          0x80000005    0x005a07 2
192.168.10.1   192.168.10.1   1465         0x8000000b    0x00d1dc 2
192.168.20.1   192.168.20.1   1465         0x8000000a    0x002d0e 3

      Net Link States (Area 0)

Link ID        ADV Router    Age          Seq#           Checksum
10.0.23.2      192.168.30.1   689          0x80000003    0x00cf24
10.0.12.2      192.168.20.1   1465         0x80000002    0x00bc8e
R3#
```


7.3 OSPF Routing Table

R1# show ip route

```
R1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.12.0/24 is directly connected, GigabitEthernet0/0
L       10.0.12.1/32 is directly connected, GigabitEthernet0/0
O       10.0.23.0/24 [110/2] via 10.0.12.2, 00:55:29, GigabitEthernet0/0
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/1
L       192.168.10.1/32 is directly connected, GigabitEthernet0/1
O       192.168.20.0/24 [110/2] via 10.0.12.2, 00:55:29, GigabitEthernet0/0
O       192.168.30.0/24 [110/3] via 10.0.12.2, 00:55:29, GigabitEthernet0/0
```

R1>

R2# show ip route

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.12.0/24 is directly connected, GigabitEthernet0/0
L       10.0.12.2/32 is directly connected, GigabitEthernet0/0
C       10.0.23.0/24 is directly connected, GigabitEthernet0/1
L       10.0.23.1/32 is directly connected, GigabitEthernet0/1
O       192.168.10.0/24 [110/2] via 10.0.12.1, 00:55:15, GigabitEthernet0/0
    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/2
L       192.168.20.1/32 is directly connected, GigabitEthernet0/2
O       192.168.30.0/24 [110/2] via 10.0.23.2, 01:12:24, GigabitEthernet0/1
```

R2#

R3# show ip route

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O       10.0.12.0/24 [110/2] via 10.0.23.1, 00:54:54, GigabitEthernet0/0
C       10.0.23.0/24 is directly connected, GigabitEthernet0/0
L       10.0.23.2/32 is directly connected, GigabitEthernet0/0
O       192.168.10.0/24 [110/3] via 10.0.23.1, 00:54:44, GigabitEthernet0/0
O       192.168.20.0/24 [110/2] via 10.0.23.1, 01:12:03, GigabitEthernet0/0
    192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/1
L       192.168.30.1/32 is directly connected, GigabitEthernet0/1
```

R3#

8. Problems Faced & Solutions Applied

Several issues were encountered during the lab. The table below summarizes each problem, its root cause, and the solution applied.

Problem	Root Cause	Solution Applied
PCs not getting DHCP IP	DHCP pool not configured / interface down	Configured ip dhcp pool; set interface to no shutdown
Ping to DHCP server failed	VLAN mismatch or switchport shutdown	Assigned switchports to VLAN20; used no shutdown
DHCP request still failing	No physical DHCP server connected	Used router itself as DHCP server with pool config
DNS ping 'PC4' fails	DNS server IP missing or no hostname mapping	Configured DNS server IP, added hostname → IP mapping, set DNS in DHCP pool
OSPF neighbors not forming	Wrong network statements / subnet mismatch	Used correct network x.x.x.x 0.0.0.255 area 0; verified interfaces up
Ping between routers fails	Wrong cable type or interface shutdown	Cross-checked R1–R2–R3 connections; issued no shutdown on all interfaces

9. Testing & Verification Commands

The following commands were used to verify the network is operating correctly. Each command and its expected result are documented below.

9.1 OSPF Verification

```
show ip ospf neighbor      ! Confirms neighbor adjacencies
show ip ospf database      ! Shows LSAs flooded in Area 0
show ip route              ! Shows OSPF routes (marked 'O')
```

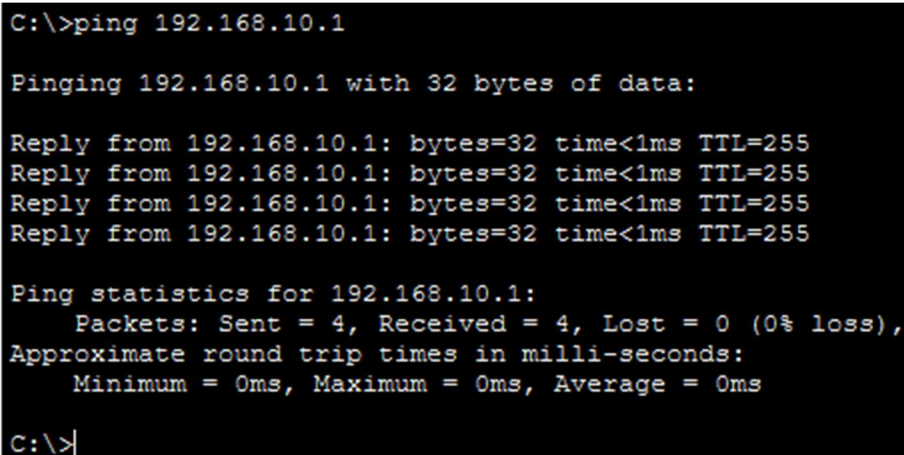
9.2 DHCP Verification

```
show ip dhcp binding      ! Shows IPs assigned to clients
show ip dhcp pool          ! Shows pool usage and statistics
```

9.3 Connectivity Tests

```
ping 192.168.10.1         ! VLAN10 gateway
ping 192.168.20.1         ! VLAN20 gateway
ping 192.168.30.1         ! VLAN30 gateway
ping PC4                   ! DNS hostname resolution test
```

Screenshot: Successful ping results across VLANs



```
C:\>ping 192.168.10.1

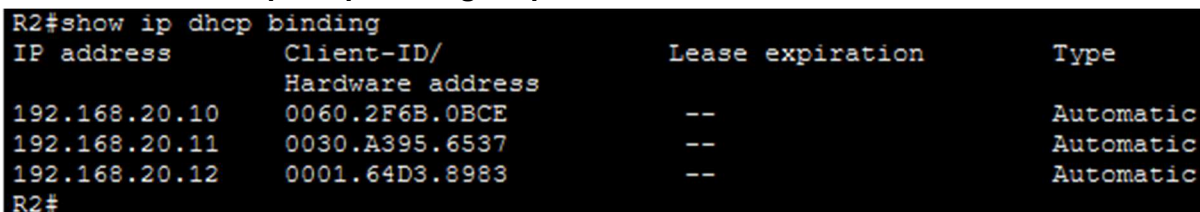
Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Screenshot: show ip dhcp binding output



```
R2#show ip dhcp binding
IP address      Client-ID/
                Hardware address    Lease expiration        Type
192.168.20.10   0060.2F6B.0BCE                    --                       Automatic
192.168.20.11   0030.A395.6537                    --                       Automatic
192.168.20.12   0001.64D3.8983                    --                       Automatic
R2#
```

Screenshot: show ip dhcp pool output**R1**

```
R1>show ip dhcp pool

Pool VLAN10 :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 3
  Excluded addresses                : 1
  Pending event                    : none

  1 subnet is currently in the pool
  Current index    IP address range    Leased/Excluded/Total
  192.168.10.1     192.168.10.1 - 192.168.10.254  3 / 1 / 254
R1>
```

R2

```
R2#show ip dhcp pool

Pool VLAN20 :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 3
  Excluded addresses                : 1
  Pending event                    : none

  1 subnet is currently in the pool
  Current index    IP address range    Leased/Excluded/Total
  192.168.20.1     192.168.20.1 - 192.168.20.254  3 / 1 / 254
```

R3

```
R3#show ip dhcp pool

Pool VLAN30 :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 3
  Excluded addresses                : 1
  Pending event                    : none

  1 subnet is currently in the pool
  Current index    IP address range    Leased/Excluded/Total
  192.168.30.1     192.168.30.1 - 192.168.30.254  3 / 1 / 254
R3#
```

10. Conclusion

This lab successfully demonstrated the configuration of a multi-site network using OSPF as the routing protocol. Three routers were connected in a linear topology with OSPF Area 0, and each site was configured with its own VLAN, DHCP pool, and default gateway.

A DNS server was deployed on the VLAN20 segment, enabling hostname-based communication between devices. Route summarization was practiced to understand how specific routes can be aggregated to optimize routing table size.

All OSPF neighbor relationships formed correctly, LSAs propagated across the network as verified by the show ip ospf database command, and end-to-end connectivity was confirmed through successful ping tests both by IP address and hostname.

The troubleshooting experience reinforced the importance of verifying interface states, VLAN assignments, and correct OSPF network statements when diagnosing network issues.